

```

1  LIBRARY ieee;
2  USE ieee.std_logic_1164.ALL;
3  USE ieee.numeric_std.ALL;
4
5  ENTITY s23_3a IS
6      PORT (
7          clk : IN STD_LOGIC;
8          u : IN STD_LOGIC;
9          x : IN STD_LOGIC_VECTOR(2 DOWNTO 0);
10         y : OUT STD_LOGIC_VECTOR(2 DOWNTO 0)
11     );
12 END ENTITY;
13
14 ARCHITECTURE behavioral OF s23_3a IS
15     SIGNAL q : STD_LOGIC_VECTOR(2 DOWNTO 0);
16     SIGNAL a, b, c, d, e, f, g, h, i, j, k, l, m, n : STD_LOGIC;
17
18 BEGIN
19
20     --flipflops
21     ff : PROCESS (clk)
22     BEGIN
23         IF rising_edge(clk) THEN
24             q(2) <= c;
25             q(1) <= j;
26             q(0) <= n;
27         END IF;
28     END PROCESS ff;
29
30     --combinational logic
31     a <= (u AND x(2));
32     b <= (NOT u) AND g;
33     c <= a OR b;
34
35     d <= q(2) AND (NOT q(1)) AND (NOT q(0));
36     e <= (NOT q(1)) AND q(0);
37     f <= (NOT q(2)) AND q(1);
38     g <= d OR e OR f;
39
40     h <= u AND x(1);
41     i <= (NOT u) AND k;
42     j <= h OR i;
43
44     k <= q(1) XOR q(0);
45
46     l <= u AND x(0);
47     m <= (NOT u) AND q(0);
48     n <= m OR l;
49
50     --output combination logic
51     y <= q;
52
53 END ARCHITECTURE behavioral;

```